AGENTR: Agentic Data Science with R

Full Program, October 2026 Cohort

Imad EL BADISY, PhD

1 Course Overview

Title: Agentic Data Science with R
Format: Online, cohort-based
Duration: 2 weeks, 8 sessions, 2 hours per session, 16 contact hours
Dates: 19/10/2026 – 01/11/2026
Days: Monday, Wednesday, Friday (18h00 to 20h00), Sunday (10h00 to 12h00)
Time: Morocco time
Price: 200€ (2000 MAD)
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1.1 Course Description

`AGENTR` is the capstone of the MHDSR pathway: a practical, hands-on introduction to *agentic data science*, using large language models (LLMs) and coding agents to build, run, and reproduce data analysis pipelines in R from the command line. It builds directly on the R foundations of the earlier courses, and a working knowledge of R is required to get full value from it.

The course works through the full agentic workflow: choosing between frontier and local models, prompting and context engineering, calling R and shell tools from an agent via structured tool use, grounding output with retrieval over data and documents, packaging reusable skills and multi-step agent workflows, and evaluating agent output with tests, guardrails, and human review.

The emphasis throughout is on keeping the analyst in control: verification, provenance, cost, privacy, and honest reporting of AI-assisted work.

1.2 Target Audience

R users, data analysts, researchers, PhD students, and public health professionals who want to use LLMs and coding agents for real analysis work. No machine learning background required.

1.3 Prerequisites

A working knowledge of R is required, along with comfort using the command line. `AGENTR` builds on the R foundations covered in `BIOSTATR`, `ML4HOR`, and `METAR`, which are the assumed baseline; prior data analysis experience is recommended but not mandatory.

1.4 Learning Objectives

By the end of `AGENTR`, participants will be able to:

1. describe what agentic systems are, and where they help or fail in data science;
2. set up a CLI-based agentic workflow for R projects (API keys, local runtimes, project scaffolding);
3. compare frontier and local LLMs on capability, cost, latency, and privacy;
4. write effective prompts and manage context for data analysis tasks;
5. call R functions and external tools from a model using structured tool use / function calling;
6. build retrieval over datasets, codebases, and documents to ground model output;
7. package reusable skills and multi-step agent workflows for recurring analyses;
8. evaluate agent output with tests and human review, and add guardrails;
9. manage reproducibility, provenance, cost, and data governance of AI-assisted analyses;
10. deliver a reproducible, agent-assisted analysis pipeline with an audit trail.

2 Detailed Schedule

Table 1: `AGENTR` detailed schedule

No.	Week	Date	Time	Topic	Main Content
1	1	19/10/2026 (Mon)	18h00-20h00	Introduction to Agentic Data Science: LLMs, Agents & the CLI Workflow for R	What LLMs and agents are, chat vs agent vs pipeline, the terminal-first workflow, project scaffolding, when agentic help pays off and when it does not
2	1	21/10/2026 (Wed)	18h00-20h00	Frontier vs Local Models: Access, Cost, Latency & Privacy	Hosted frontier APIs, local model runtimes, model families and sizes, context windows, token cost and latency, data-privacy trade-offs, choosing a model per task
3	1	23/10/2026 (Fri)	18h00-20h00	Prompting and Context Engineering for Data Analysis	Task framing, few-shot examples, structured output, supplying data dictionaries and code context, context limits, prompt iteration and versioning

No.	Week	Date	Time	Topic	Main Content
4	1	25/10/2026 (Sun)	10h00-12h00	Tool Use and Function Calling: Driving R and Shell Tools from an Agent	Structured tool/function calling, exposing R functions and shell commands as tools, the read-eval-act loop, sandboxing, approval gates, failure handling
5	2	26/10/2026 (Mon)	18h00-20h00	Retrieval-Augmented Workflows over Data, Code & Literature	Chunking and embeddings, vector stores, retrieval over project files and datasets, grounding answers in sources, citations, keeping retrieval reproducible
6	2	28/10/2026 (Wed)	18h00-20h00	Agent Skills and Reusable Multi-Step Pipelines from the Command Line	Packaging recurring analyses as skills, multi-step and multi-agent workflows, parameterized runs, scheduling, chaining agents into a data pipeline
7	2	30/10/2026 (Fri)	18h00-20h00	Evaluation, Guardrails, Provenance & Reproducibility of Agent-Assisted Work	Test-based checks on agent output, human review, guardrails and policies, logging prompts and responses, cost tracking, provenance and audit trails, honest reporting
8	2	01/11/2026 (Sun)	10h00-12h00	End-to-End Agentic Analysis Project	Full workflow: scaffold, retrieve, prompt, run tools, evaluate, and report a reproducible agent-assisted analysis with an audit trail

3 Course Materials

- Slides and self-contained notebooks distributed before each session
- R scripts, datasets, prompts, and exercises (shared repository)
- Exercises reviewed and corrected during sessions

4 Assessment

Each course ends with a short reproducible Quarto report or mini-project, presented and discussed in Session 8. For `AGENTR`, this is a reproducible agent-assisted analysis pipeline with an audit trail.

5 About the Instructor

Imad EL BADISY, PhD, is a biostatistician working on statistical and machine learning methods for health data, with a focus on survival analysis. Full profile, publications, and R packages: elbadisyimad.com

6 Policies

Language: Teaching in French, with English for statistical and methodological terminology.
Software: R ($\geq$ 4.4) and RStudio required; a terminal, an LLM CLI / coding agent, and API access and/or a local model runtime are used throughout.
Attendance: Full attendance at all 8 sessions is recommended given the cumulative structure.